Due Date

You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Wednesday, September 17, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Lab Exercise

In this exercise you are to write a simple Java program to calculate the number of days in a given month.

Find the Number of Days in a Month

Write a program that prompts the user to enter the month and year and displays the number of days in the month. For example, if the user entered month **2** and year **2012**, the program should display

```
February 2012 has 29 days.
```

If the user entered the month **3** and year **2015**, the program should display

```
March 2015 has 31 days.
```

Obviously, the tricky case in this program is for the input month value of **2** (for February): normally, February has 28 days, but in *Leap Years*, February has 29 days. **Section 3.11** of the textbook is a case study of determining if a particular year is a leap year. Be sure to try the following two cases:

```
February 2000   has   ??   days.

February 2100   has   ??   days.
```

Demonstrate your program to the instructor in class.

Sample Output

In the sample output shown below, and on the next page, text that the user types is shown in **BOLD** font. When the program actually runs, all text is shown in the same font.

Sample Input / Output
Enter a month in the year (e.g., 1 for Jan): **3** Enter a year: **2005** March 2005 has 31 days

Sample Input / Output

```
Enter a month in the year (e.g., 1 for Jan): 4
Enter a year: 1999
April 1999 has 30 days
```

Sample Input / Output

```
Enter a month in the year (e.g., 1 for Jan): 2
Enter a year: 2009
February 2009 has 28 days
```

Sample Input / Output

```
Enter a month in the year (e.g., 1 for Jan): 2
Enter a year: 2004
February 2004 has 29 days
```

Submit Solution to *Moodle* and Demonstrate Solution in Class

Submit the **source code** (*.java) file for your solution to *Moodle* as soon as you feel it is ready. Remember to follow the complete submission process. That is, follow the steps in the **Submit_Assignment_to_Moodle** document in the **Handouts** section of *Moodle*. The final *Moodle* submission status must be: **Submitted for gradng**:

Submission status	Submitted for grading

Demonstrate your solution during class, at the earliest opportunity.

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